

Experiment 10

Problem Statement:

Text analytics for Business Insights: Develop models using NLP frameworks such as spaCy, BERT, or RoBERTa for analyzing customer reviews/comments. Implement advanced text analytics techniques like Aspect-based Sentiment analysis, and Topic modelling using LDA or BERTopic

Theory:

NLP helps you extract insights from unstructured text and has many use cases, such as:

- Automatic summarization
- Named-entity recognition
- Question answering systems
- Sentiment analysis

spaCy is a free, open-source library for NLP in Python written in Cython. It's a modern, production-focused NLP library that emphasizes speed, streamlined workflows, and robust pretrained models. spaCy is designed to make it easy to build systems for information extraction or general-purpose natural language processing.

What is BERT?

BERT (Bidirectional Encoder Representations from Transformers) leverages a transformer-based neural network to understand and generate human-like language. BERT employs an encoder-only architecture. In the original Transformer architecture, there are both encoder and decoder modules. The decision to use an encoder-only architecture in BERT suggests a primary emphasis on understanding input sequences rather than generating output sequences.

Bidirectional Approach of BERT

Traditional language models process text sequentially, either from left to right or right to left. This method limits the model's awareness to the immediate context preceding the target word. BERT uses a bi-directional approach considering both the left and right context of words in a sentence, instead of analyzing the text sequentially, BERT looks at all the words in a sentence simultaneously.

ROBERTa

RoBERTa (short for “Robustly Optimized BERT Approach”) is a variant of the BERT (Bidirectional Encoder Representations from Transformers) model, which was developed by researchers at Facebook AI. Like BERT, RoBERTa is a transformer-based language model that uses self-attention to process input sequences and generate contextualized representations of words in a sentence.

One key difference between RoBERTa and BERT is that RoBERTa was trained on a much larger dataset and using a more effective training procedure. In particular, RoBERTa was trained on a dataset of 160GB of text, which is more than 10 times larger than the dataset used to train BERT. Additionally, RoBERTa uses a dynamic masking technique during training that helps the model learn more robust and generalizable representations of words.

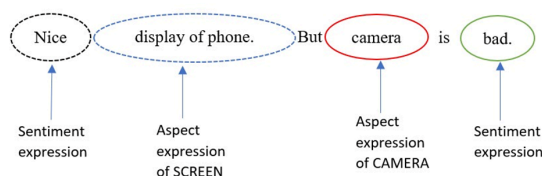
RoBERTa has been shown to outperform BERT and other state-of-the-art models on a variety of natural language processing tasks, including language translation, text classification, and question answering. It has also been used as a base model for many other successful NLP models and has become a popular choice for research and industry applications.

Overall, RoBERTa is a powerful and effective language model that has made significant contributions to the field of NLP and has helped to drive progress in a wide range of applications.

RoBERTa stands for Robustly Optimized BERT Pre-training Approach. It was presented by researchers at Facebook and Washington University. The goal of this paper was to optimize the training of BERT architecture in order to take less time during pre-training.

Aspect-based Sentiment analysis

Before we get into the several strategies for aspect-level sentiment analysis, we must first define an **aspect**. Aspects are a set of semantically rich, concept-centric terms that signify specific features or characteristics of the review or comment. For example, If we consider Example shown below Location, and food are all aspects to consider.

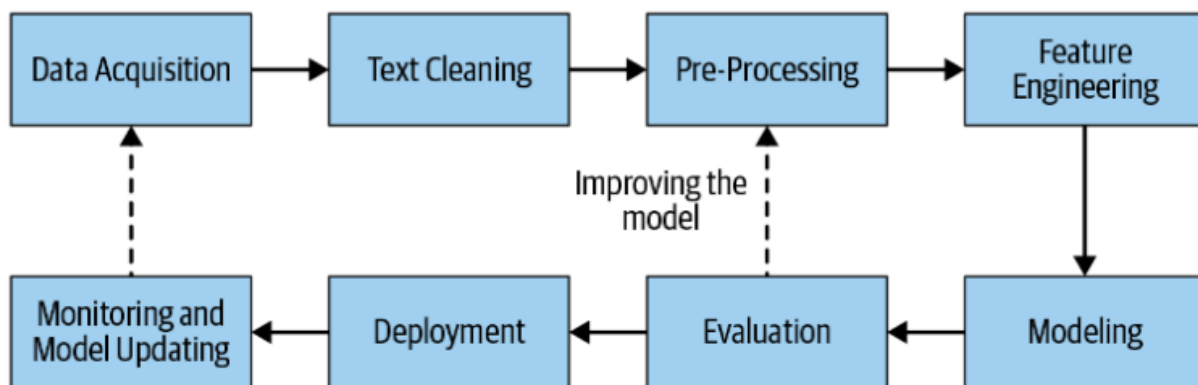


Aspect-based sentiment analysis (ABSA) is a text analysis technique that categorizes data by aspect and identifies the sentiment attributed to each one. We will use it to analyze customer feedback for flipkart products by associating specific sentiments with different aspects of a product or service.

Aspect Based Sentiment Analysis can be categorized into two types :-

- **Supervised ABSA** :- In the Supervised approach we provide the seed words which then help the supervised model to extract aspects of the sentence. If it identifies the particular aspects for that seed then it tags those sentences with those aspects.
- **Unsupervised ABSA** :- As it is obvious providing the seed for each sentence is a difficult task plus let say if our dataset is large (20-100k) then that will become a nightmare. Topic Modeling is the technique which helps extract latent topics from a document, in our case we will extract aspects from a sentence using topic modeling techniques such as latent Dirichlet algorithm (LDA).

Aspect Based Sentiment Analysis is a Natural Language Processing (NLP) Based Approach so we will deal with it like that and follow the below mentioned pipeline for a NLP task.



What is Topic Modeling?

Topic modeling is a type of statistical modeling used to uncover the abstract topics that occur in a collection of documents. It is a form of unsupervised learning, which means it does not require labeled data. Instead, it relies on the co-occurrence patterns of words within the documents to discover latent topics.

Key Concepts in Topic Modeling

1. Corpus: A large collection of text documents.
2. Topics: Abstract themes or concepts that are represented by clusters of words that frequently appear together.
3. Topic Distribution: The distribution of topics within a document, indicating the presence and prominence of various topics.
4. Word Distribution: The distribution of words within a topic, highlighting the most representative words for each topic.

What is Latent Dirichlet Allocation (LDA)?

Latent Dirichlet Allocation (LDA) is a generative probabilistic model designed to discover latent topics in large collections of text documents. Introduced by David Blei, Andrew Ng, and Michael Jordan in 2003, LDA assumes that each document is a mixture of topics and that each topic is a mixture of words. The goal of LDA is to identify these topics and determine the distribution of topics within each document and the distribution of words within each topic.

How LDA Works

LDA operates on the following principles:

- Document-Topic Distribution: Each document is represented as a distribution over topics. This means that a document can belong to multiple topics, each with a certain probability.
- Topic-Word Distribution: Each topic is represented as a distribution over words. This implies that a topic is defined by a set of words, each associated with a probability of appearing in that topic.
- Generative Process: LDA uses a generative process to create documents. This involves:
 1. Choosing a distribution of topics for each document.
 2. For each word in the document, selecting a topic based on the topic distribution.
 3. Generating the word from the selected topic's word distribution.